Lecture 35 — Eigenvectors as persistent directions

Mon Nov 16

Last updated: August 18, 2026.

The class outline for this day will be posted before class.

Reading: beginning of Strang §6.1. Related objectives: Lectures.

Learning objectives

By the end of class, students should be able to:

1. Explain geometrically what makes v an eigenvector.
2. Verify a proposed eigenpair.
3. Find λ when an eigenvector is given.
4. Connect eigenvectors to repeated transformations and steady directions.
5. Recognize that a rotation may have no real eigenvector, motivating complex numbers.